

1 Quantum and Classical Statistics

1.1 A very short summary of quantum mechanics

In many cases we will assume that the Hamiltonian has the form

$$H = \frac{p^2}{2m} + V(x). \quad (1)$$

This definition holds both for classical and quantum mechanics, although we typically put a hat on operators in quantum mechanics. x and p are vectors of f dimensions, where in many situations f is 3 times the number of atoms and $V(x)$ is a scalar field of these variables.

Eigenstates are defined by the time-independent Schrödinger equation:

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle \quad (2)$$

but remember that wavefunctions do not need to be eigenstates (but can be defined as linear combinations of eigenstates).

Consider a system (e.g. a single molecule) described by the state $|\psi\rangle$. An expectation value of an operator, $\hat{A}$, is given by

$$\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle. \quad (3)$$

If $\hat{A} = A(\hat{x})$, this can be evaluated in the position representation by inserting a complete set of position states $1 = \int |x\rangle \langle x| dx$ to give

$$\langle \hat{A} \rangle_\psi = \int \langle \psi | \hat{A} | x \rangle \langle x | \psi \rangle dx \quad (4a)$$

$$= \int \langle \psi | x \rangle A(x) \langle x | \psi \rangle dx \quad (4b)$$

$$= \int \psi(x)^* A(x) \psi(x) dx, \quad (4c)$$

which is the formula you may be more familiar with. For the specific example of finding the probability of finding the particle at a point $x = a$, use $\hat{A} = \delta(\hat{x} - a)$ to give

$$\langle \delta(\hat{x}_a) \rangle_\psi = \int \psi(x)^* \delta(x - a) \psi(x) dx \quad (5a)$$

$$= |\psi(a)|^2. \quad (5b)$$

1.2 Quantum statistical ensembles

If we do a measurement on a large number of molecules, the observable will be an average over all the molecules, leading to

$$\langle \hat{A} \rangle = \sum_n P_n \langle \psi_n | \hat{A} | \psi_n \rangle, \quad (6)$$

where P_n is the probability that a molecule chosen at random is in state ψ_n and we sum over all possible states $|\psi_n\rangle$. The ensemble of molecules is thus defined by the probabilities.

We define the density operator

$$\hat{\rho} = \sum_n P_n |\psi_n\rangle\langle\psi_n|, \quad (7)$$

and note that we can obtain the same ensemble average using

$$\langle\hat{A}\rangle_\rho = \text{Tr} [\hat{\rho} \hat{A}], \quad (8)$$

where the quantum trace is defined analogously to the matrix trace as a sum over diagonal elements

$$\text{Tr} [\hat{O}] = \sum_n \langle\psi_n|\hat{O}|\psi_n\rangle. \quad (9)$$

Using these definitions, it is easy to show [make sure you can do this]

$$\text{Tr} [\hat{\rho}] = \sum_{n,m} P_m \langle\psi_n|\psi_m\rangle \langle\psi_m|\psi_n\rangle = \sum_n P_n = 1, \quad (10)$$

where we have used the fact that the wavefunctions are orthonormal and that the probabilities sum to 1. This is the normalization condition for the density operator.

To find the population of a particular state, use the projection operator $\hat{A} = |\psi_m\rangle\langle\psi_m|$. This gives

$$\langle\hat{A}\rangle_\rho = \text{Tr} [\hat{\rho} |\psi_m\rangle\langle\psi_m|] = \langle\psi_m|\hat{\rho}|\psi_m\rangle = P_m. \quad (11a)$$

Also note the properties of the trace. It is invariant to cyclic permutation

$$\text{Tr}[ABC] = \text{Tr}[CAB] = \text{Tr}[BCA]$$

and to unitary transformation ($S^\dagger = S^{-1}$):

$$\text{Tr}[S^\dagger AS] = \text{Tr}[A].$$

Because of these properties, we can find even more general definitions of the density operator which are unitary transforms of the form given above. The expression can be rewritten in an arbitrary basis set, $\{|\phi_i\rangle\}$ by using the completeness of the states, $1 = \sum_i |\phi_i\rangle\langle\phi_i|$. Then the expectation value can be then written as

$$\langle\hat{A}\rangle_\rho = \sum_n P_n \langle\psi_n|\hat{A}|\psi_n\rangle \quad (12a)$$

$$= \sum_{n,i,j} P_n \langle\psi_n|\phi_j\rangle \langle\phi_j|\hat{A}|\phi_i\rangle \langle\phi_i|\psi_n\rangle \quad (12b)$$

$$= \sum_{n,i,j} \langle\phi_i|\psi_n\rangle P_n \langle\psi_n|\phi_j\rangle A_{ji} \quad (12c)$$

$$= \sum_{i,j} \langle\phi_i|\hat{\rho}|\phi_j\rangle A_{ji} \quad (12d)$$

$$= \sum_{i,j} \rho_{ij} A_{ji} = \text{Tr} [\hat{\rho} \hat{A}]. \quad (12e)$$

In the final line we have defined the density matrix with elements $\rho_{ij} = \langle \phi_i | \hat{\rho} | \phi_j \rangle$ and note that the sums are equivalent to a standard matrix product and trace.

If all molecules are in the same state then one of the probabilities will be 1 and all others 0, leading to

$$\hat{\rho} = |\psi\rangle\langle\psi|.$$

This is called a pure state, as opposed to a mixed state. To determine the purity, compute

$$\text{Tr}[\hat{\rho}^2],$$

which is clearly equal to 1 only for a pure state.

1.3 Classical statistical ensembles

In quantum mechanics the state of the system is completely determined by the wavefunction. In classical mechanics one specifies the positions and momenta of all particles. The classical equivalent of the density operator is the probability density function $\rho(x, p)$. This should be normalized over all phase space such that

$$(2\pi\hbar)^{-f} \iint \rho(x, p) dx dp = 1. \quad (13)$$

The average of an observable $A \equiv A(x, p)$ in a particular distribution ρ is

$$\langle A \rangle_\rho^{\text{cl}} = (2\pi\hbar)^{-f} \iint \rho(x, p) A(x, p) dx dp, \quad (14)$$

where f are the number of degrees of freedom (typically 3 times the number of atoms).

One thus sees that the classical analogue of the trace is an integral over phase space. The prefactor is required to keep the units consistent and its value related to Planck's constant is determined to ensure quantum-classical correspondence, as we will show later.

1.4 Wigner functions

The Wigner distribution function is defined as

$$\rho_W(x, p) = \int \langle x + \frac{1}{2}\xi | \hat{\rho} | x - \frac{1}{2}\xi \rangle e^{-ip\xi/\hbar} d\xi. \quad (15)$$

Note the normalization such that $(2\pi\hbar)^{-f} \iint \rho_W(x, p) dx dp = 1$.

One can further show that [check that you can do this: Hint, use property of delta function $(2\pi\hbar)^{-f} \int e^{\pm ip \cdot x/\hbar} dp = \delta(x)$]

$$\langle \hat{A} \rangle_\rho = \text{Tr} [\hat{\rho} \hat{A}] = (2\pi\hbar)^{-f} \iint \rho_W(x, p) A_W(x, p) dx dp, \quad (16)$$

where $A_W(x, p)$ is defined in terms of $\hat{A}$ in the same way as equation (15). It is easy to show that if it is an operator of position only, then this is equivalent to simply removing the hats! In this way, we can compute quantum-mechanical ensemble averages using a classical-type method.

For example, the ground-state harmonic oscillator wavefunction is

$$\psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2},$$

where $\alpha = m\omega/\hbar$, and the first excited state is

$$\psi_1(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2\alpha} x e^{-\alpha x^2/2}.$$

In either case, we use $\hat{\rho} = |\psi_n\rangle\langle\psi_n|$ to get

$$\rho_W(x, p) = 2e^{-\frac{p^2}{m\omega\hbar} - \frac{m\omega x^2}{\hbar}}$$

for the ground state and

$$\rho_W(x, p) = 2 \left(\frac{2p^2}{m\omega\hbar} + \frac{2m\omega x^2}{\hbar} - 1 \right) e^{-\frac{p^2}{m\omega\hbar} - \frac{m\omega x^2}{\hbar}}$$

for the excited state.

1.5 Thermal equilibrium and partition functions

Statistical properties are average values of an equilibrium distribution and are therefore not time dependent. The partition function is one of the most important quantities as other thermodynamic properties can be derived from it. Throughout we use $\beta = (k_B T)^{-1}$.

1.5.1 Thermal density matrix

The density operator which describes thermal equilibrium is the Boltzmann operator

$$\hat{\rho}_{\text{eq}} = Q^{-1} e^{-\beta \hat{H}}, \quad (17)$$

where this is the exponential of a matrix [not the same as a matrix of exponentials!] defined by

$$e^{\hat{A}} = \sum_{k=0}^{\infty} \frac{1}{k!} \hat{A}^k. \quad (18)$$

We require $\text{Tr}[\hat{\rho}] = 1$ which implies that the normalization factor is

$$Q = \text{Tr} \left[e^{-\beta \hat{H}} \right]. \quad (19)$$

This is the most general expression for the quantum partition function.

We can evaluate the trace in the basis of eigenstates of the Hamiltonian, $|\psi_n\rangle$, to give

$$\begin{aligned} Q &= \sum_n \langle \psi_n | e^{-\beta \hat{H}} | \psi_n \rangle \\ &= \sum_n \langle \psi_n | e^{-\beta E_n} | \psi_n \rangle \\ &= \sum_n e^{-\beta E_n}. \end{aligned}$$

Here we used the Taylor series expression for the matrix exponential, equation (18), as well as the eigenvalue relationship, $\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle$.

1.5.2 Classical density

The probability density in classical mechanics is $\rho(x, p)$, for instance $\rho_{\text{eq}}(x, p) = e^{-\beta H(x, p)} / Q_{\text{cl}}$ is the Boltzmann distribution.

In many cases one cannot actually compute the normalization factor Q_{cl} , but can nonetheless obtain equilibrium averages from thermostatted molecular dynamics simulations or Metropolis Monte Carlo (see further reading on Molecular Simulation):

$$\langle A \rangle_{\text{eq}}^{\text{cl}} = \frac{\iint e^{-\beta H(x, p)} A(x, p) dx dp}{\iint e^{-\beta H(x, p)} dx dp}. \quad (20)$$

This expression should be interpreted as taking an average of $A(x, p)$ from an unnormalized distribution $e^{-\beta H}$.

1.5.3 Partition functions

The quantum partition function can be defined as

$$Q = \sum_n e^{-\beta E_n}, \quad (21)$$

where E_n are the eigenvalues of the Hamiltonian. This formula is therefore only useful if the eigenvalues are known or can be computed.

The classical partition function can be defined as

$$Q_{\text{cl}} = (2\pi\hbar)^{-f} \iint e^{-\beta H(x, p)} dx dp, \quad (22)$$

where f is the number of degrees of freedom. The momentum integral is a Gaussian so can be performed explicitly to give

$$Q_{\text{cl}} = \left(\frac{m}{2\pi\beta\hbar^2} \right)^{f/2} \int e^{-\beta V(x)} dx. \quad (23)$$

1.5.4 Example: Free particle

The free particle potential is given by $V(x) = 0$. Therefore the classical partition function for a free particle in one degree of freedom is

$$Q_{\text{cl}} = \lim_{L \rightarrow \infty} \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2} \int_{-L/2}^{L/2} dx \quad (24a)$$

$$= \lim_{L \rightarrow \infty} \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2} L \quad (24b)$$

$$Q_{\text{cl}}/L = \left(\frac{m}{2\pi\beta\hbar^2} \right)^{1/2}. \quad (24c)$$

Therefore for a free particle, it only makes sense to talk about the partition-function per unit length. For an atom or molecule in three-dimensional space, the translational partition function is $(m/2\pi\beta\hbar^2)^{3/2}$ per unit volume.

For the quantum version, we will use more sophisticated techniques introduced later in the course. However, note now that the result is exactly equivalent to the classical version in this case. For an alternative derivation, use the states of a particle in a box of length L and then take the limit $L \rightarrow \infty$.

1.5.5 Example: Harmonic Oscillator

The one-dimensional harmonic oscillator potential is given by $V(x) = \frac{1}{2}m\omega^2x^2$. Therefore the classical partition function is

$$Q_{\text{cl}} = \frac{1}{\beta\hbar\omega} \quad \text{for a 1D harmonic oscillator.} \quad (25)$$

The quantum energy levels are $E_n = (n + \frac{1}{2})\hbar\omega$ for $n \in \{0, 1, 2, \dots\}$ so that

$$Q = \sum_{n=0}^{\infty} e^{-\beta(n+\frac{1}{2})\hbar\omega} \quad (26a)$$

$$= e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n \quad (\text{N.B. geometric series}) \quad (26b)$$

$$= \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \quad (26c)$$

$$= [2 \sinh \beta\hbar\omega/2]^{-1}. \quad (26d)$$

1.5.6 Example: rotational partition function

The moment of inertia for a *linear* system is $I = \sum_i m_i r_i^2$ where r_i is the distance from the centre of mass to the atom i . The rotational constant is then given by $B = \hbar^2/2I$.

For a linear system, the energy levels are $E_J = BJ(J+1)$ with degeneracy $2J+1$. The quantum partition function is thus given by

$$Q = \sum_{J=0}^{\infty} (2J+1) e^{-\beta BJ(J+1)} \quad \text{for a linear rotor.} \quad (27)$$

The classical partition function can be found by approximating this sum by an integral to give

$$Q_{\text{cl}} = \frac{1}{\beta B} \quad \text{for a linear rotor.} \quad (28)$$

For a nonlinear rotor, the energy levels are more complicated. However, the classical partition function is known to be

$$Q_{\text{cl}} = \sqrt{\frac{\pi}{\beta^3 ABC}} \quad \text{for a nonlinear rotor,} \quad (29)$$

where A , B and C are the rotational constants defined as B above for the three principal moments of inertia.

1.5.7 Example: average energy

The average energy of a quantum system in thermal equilibrium is

$$\langle E \rangle = \text{Tr} [\hat{\rho}_{\text{eq}} \hat{H}] \quad (30a)$$

$$= Q^{-1} \sum_n E_n e^{-\beta E_n}, \quad (30b)$$

where E_n are the eigenvalues of the Hamiltonian.

For a classical system,

$$\langle E \rangle^{\text{cl}} = (2\pi\hbar)^{-f} \iint Q^{-1} e^{-\beta H(x,p)} H(x,p) \, dx \, dp \quad (31)$$

In both cases, the average energy can be defined directly from the partition function:

$$\langle E \rangle = -\frac{d}{d\beta} \ln Q. \quad (32)$$

1.6 Transition-state theory

Eyring TST is normally defined as

$$k_{\text{TST}}(T) = \frac{k_{\text{B}}T}{h} \frac{Q^\ddagger(T)}{Q_{\text{r}}(T)} e^{-\beta V^\ddagger}, \quad (33)$$

where $Q_{\text{r}}(T)$ and $Q^\ddagger(T)$ are the partition functions of the reactants and transition state (per unit length/volume etc. as appropriate) and $V^\ddagger$ is the barrier height. One typically separates the vibrations and rotations using the rigid-rotor approximation and uses quantum harmonic-oscillator partition functions for the vibrational modes and classical partition functions for the rotations. One degree of freedom of the transition state corresponds to the reaction coordinate and is ignored in this theory. We will show where this formula comes from later in the course.

1.7 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Basic Quantum mechanics and bra-ket notation

- Dirac, Quantum Mechanics (Oxford, 1948)
- Griffiths & Schroeter, Introduction to Quantum Mechanics (Cambridge), chapter 3

Statistical ensembles

- Tuckerman, Statistical Mechanics (Oxford), chapters 4.1-4.5 and 10

Molecular Simulations

- Tuckerman, Statistical Mechanics (Oxford)

- Frenkel and Smit, Understanding Molecular Simulations (Academic Press)

Wigner functions

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 5
- Schleich, Quantum Optics in Phase Space (Wiley), chapter 3

Partition functions

- Atkins, Physical Chemistry (Oxford), chapter 15
- McQuarrie, Statistical Mechanics

Transition-state theory

- Atkins, Physical Chemistry (Oxford), section 21C